Evaluating Conditional Image Generation and Classifier-Free Guidance in Diffusion Models

Jaden Vo
j9vo@ucsd.edu

Yusu Wang
email@ucsd.edu

Abstract

Diffusion models have demonstrated strong capabilities in generating realistic images, but their performance can vary depending on conditioning methods, guidance strength, and the amount of training data available. This project investigates these factors through a series of experiments using Elucidating the Design Space of Diffusion-Based Generative Models (EDM) and classifier-free guidance (CFG).

Experiments are conducted using two image datasets, CIFAR-10 and MNIST. Pretrained EDM models are used to generate unconditional and class-conditioned CIFAR-10 images, while classifier-free guidance is evaluated across multiple guidance weights. Image generation performance is measured using classification accuracy, Fréchet Inception Distance (FID), and class-specific FID. Additionally, smaller EDM networks are trained on subsets of CIFAR-10 containing 100 and 1,000 images to investigate training behavior and memorization. A separate conditional generative model is trained on MNIST to evaluate classifier-free guidance on a different image dataset.

Preliminary results on CIFAR-10 demonstrate that increasing guidance strength substantially improves class accuracy, reaching 99.5% at a guidance weight of 3. However, overall FID increases with stronger guidance, suggesting a tradeoff between class adherence and image diversity. The remaining experiments investigate how training data availability and guidance influence generative performance across different datasets.

Website: https://website.com
Code: https://github.com/javomode/diffusion-models-cifar10-mnist

1 Introduction . . . 3
2 Methods . . . 4
3 Results . . . 13
4 Discussion . . . 13
5 Conclusion . . . 13
References . . . 13

Appendices A1

1 Introduction

Diffusion models generate images by learning to reverse a gradual noising process. A denoising network estimates the clean image from a noisy one, and a sampler applies it repeatedly, moving from pure noise to a finished sample. Sample quality depends on choices beyond the architecture: whether the model sees class labels, how strongly conditioning is applied at sampling time, and how much data the model is trained on.

This project studies three questions on small image datasets, where every experiment can be run and measured end to end. First, what does class conditioning add over an unconditional model? Second, how does classifier-free guidance (CFG) trade class fidelity against realism and diversity as the guidance weight w increases? Third, what happens when the network is trained on very small subsets of the data, where memorization is expected?

1.1 Related Work

Diffusion models generate images by learning to reverse a noising process (Sohl-Dickstein et al. 2015; Ho, Jain and Abbeel 2020; Song et al. 2021). Karras et al. (2022) break this into separate design choices (network preconditioning, noise schedule, sampler, and loss weighting). The resulting EDM models reach FID 1.97 (unconditional) and 1.79 (conditional) on CIFAR-10, and we use them throughout.

Generation can be steered toward a class by classifier guidance (Dhariwal and Nichol 2021), or without a classifier by classifier-free guidance (CFG) (Ho and Salimans 2022). More recent work from our course steers unconditional models at inference time without retraining (Wang, Belkin and Wang 2026), and we use CFG as the baseline for controlling generation.

We measure quality with the Fréchet Inception Distance (FID) (Heusel et al. 2017), which depends on sample size, so we report the number of images with every value. Finally, diffusion models can reproduce training images when data is scarce (Carlini et al. 2023), which motivates our nearest-training-image comparison.

1.2 Data

CIFAR-10 (Krizhevsky, 2009) contains 60,000 color images of size 32×32 in 10 classes, split into 50,000 training and 10,000 test images. We use it to sample from NVIDIA's pretrained EDM models, to run the CFG sweep, and to train on balanced subsets of 100 and 1,000 images.

Label accuracy is measured with a pretrained ResNet-56 that only scores images and never guides the sampler. FID is computed with EDM's Inception network against reference statistics from the 50,000 training images.

MNIST (LeCun et al., 1998) contains 60,000 training and 10,000 test grayscale digit images of size 28×28 in 10 classes. We train a conditional U-Net from scratch on the 60,000

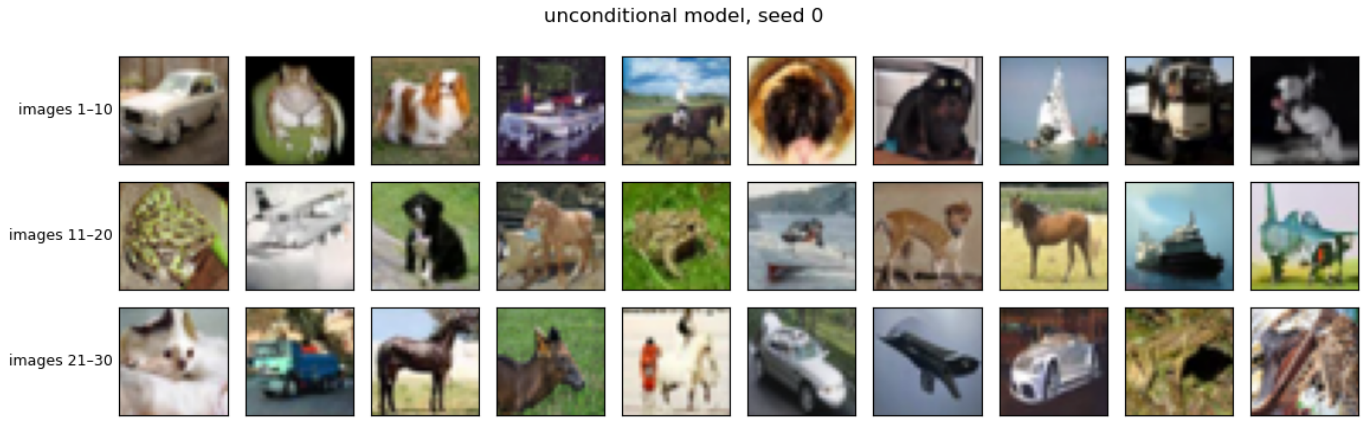


Figure 1: Thirty images generated by the pretrained unconditional EDM model for CIFAR-10 (seed 0), using EDM's deterministic Heun sampler with 18 noise levels (35 network evaluations per image).

training digits, replacing the label with a "no label" token for a random 10% of examples, and evaluate generated digits with a frozen digit classifier.

2 Methods

On CIFAR-10 we apply CFG with two pretrained networks, because EDM's released conditional network has no "no label" input, so the unconditional estimate comes from the separate unconditional network.

For our own training runs, on CIFAR-10 subsets and on MNIST, we use a single network with label dropout.

Our experiments consist of four parts:

1. Sample from the unconditional and conditional models and examine the sampler's trajectory.
2. Sweep $w \in \{0, 1, 1.5, 2, 3, 4\}$ at 1,000 images per setting.
3. Train small EDM networks on 100 and 1,000 images and track FID across snapshots.
4. Train an MNIST CFG network and sweep $w \in \{0, 1, 2, 4\}$.

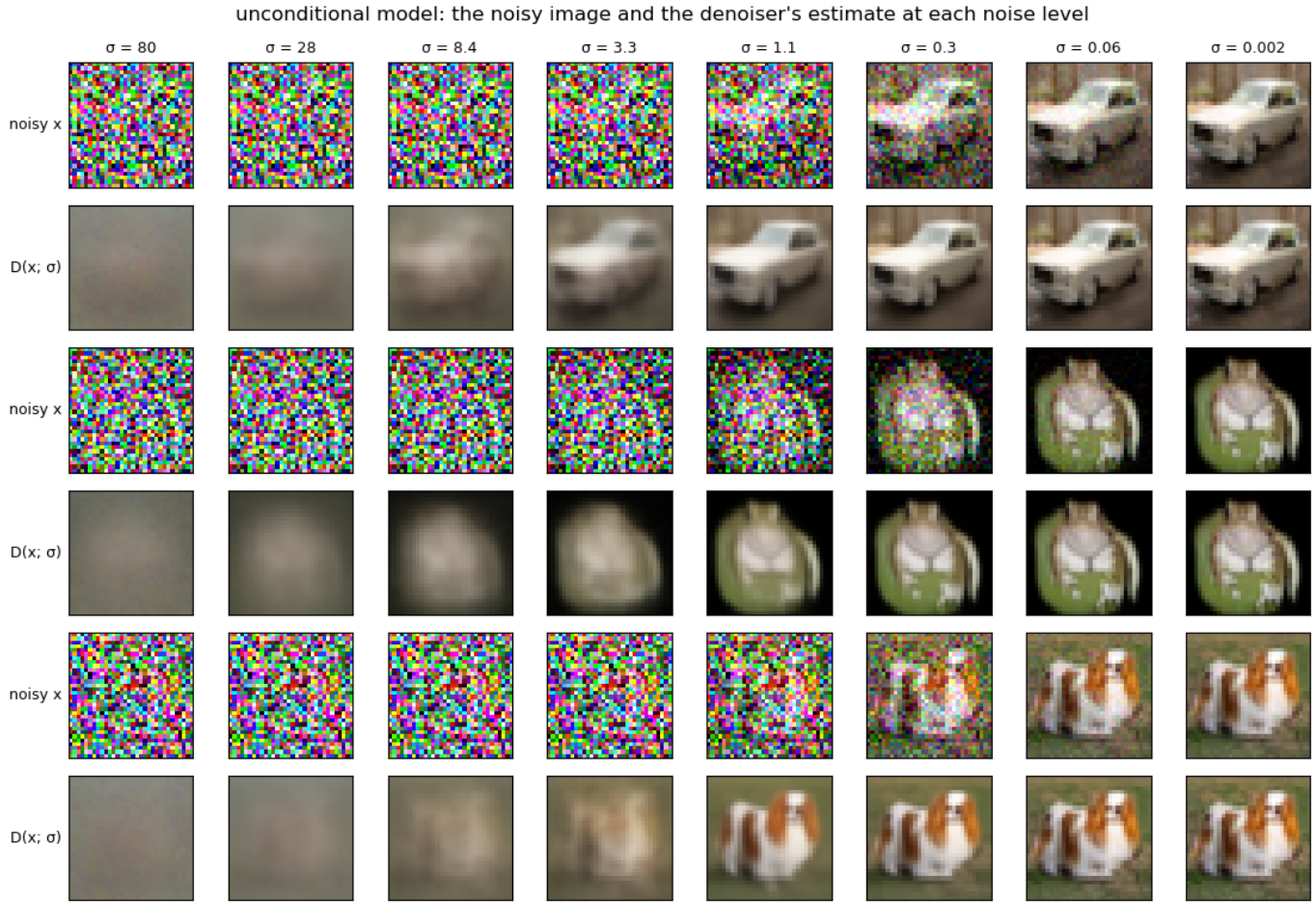


Figure 2: Sampling trajectory of the unconditional model for three images (seed 0). For each image, the upper row is the noisy image x (displayed as $x/\sqrt{1+\sigma^2}$) and the lower row is the denoiser's clean-image estimate $D(x;\sigma)$, at $\sigma \in \{80, 28, 8.4, 3.3, 1.1, 0.3, 0.06, 0.002\}$.

Table 1: CFG sweep with the two pretrained EDM networks, using 1,000 generated images per w with balanced requested classes (100 per class). Accuracy is measured by a ResNet-56 classifier. FID compares all 1,000 images with the 50,000 CIFAR-10 training images. $\text{FID}_{\text{class}}$ compares the 100 images of each class with that class's 5,000 real training images and averages over the ten classes; it is undefined at $w = 0$ because that setting ignores the requested class. FIDs from 1,000 samples are much higher than published values and are comparable only with each other.

w	Label accuracy (%)	FID	$\text{FID}_{\text{class}}$
0	11.5	31.19	–
1	95.5	31.33	112.12
1.5	98.7	31.44	107.62
2	98.8	33.05	108.89
3	99.5	36.03	112.45
4	98.9	39.03	117.53

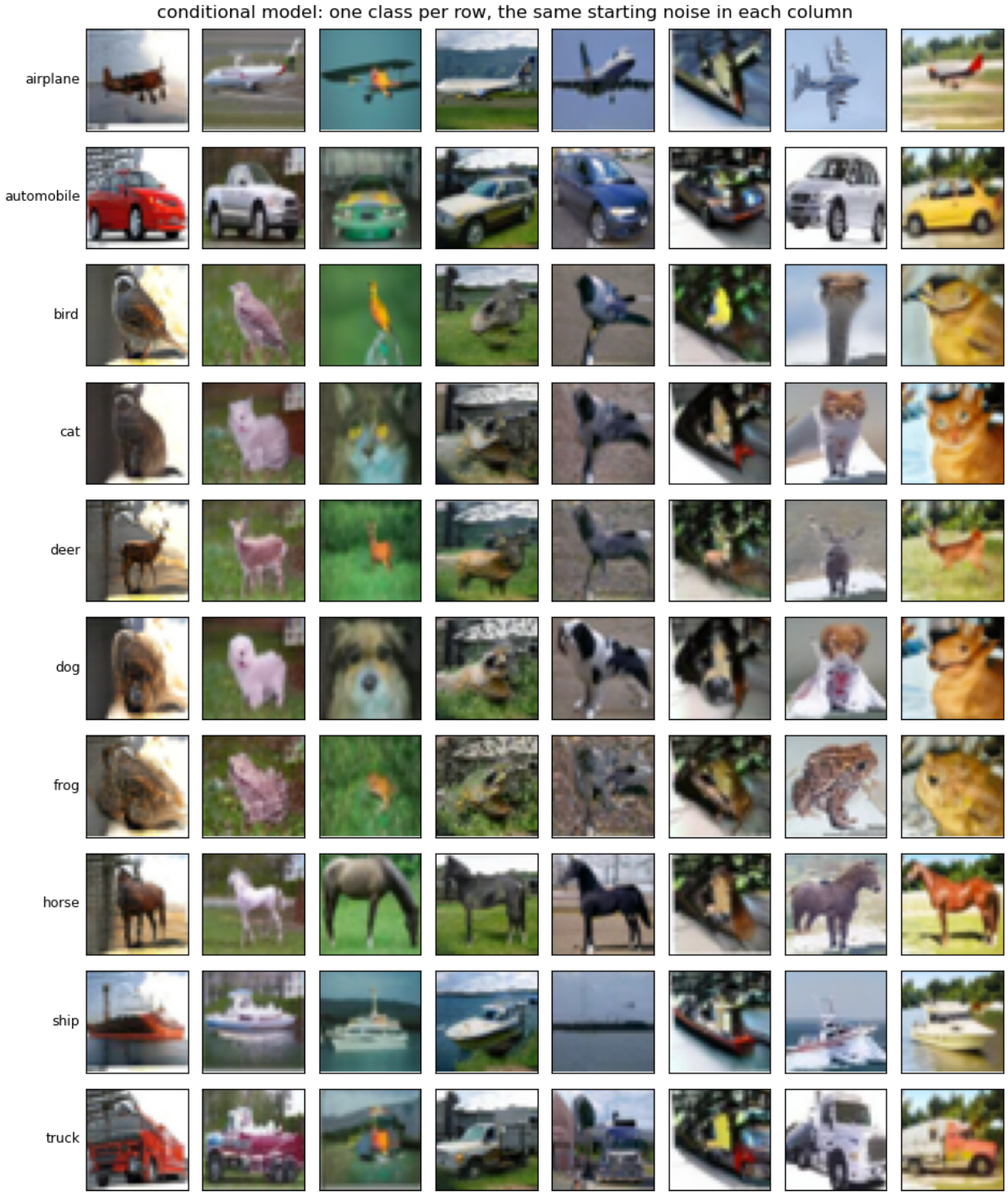


Figure 3: Eighty images from the pretrained conditional EDM model (seed 1), with one CIFAR-10 class per row and eight images per class. Each column starts from the same noise, so images in a column share layout and color cues (for example the background and lighting) even though their classes differ.

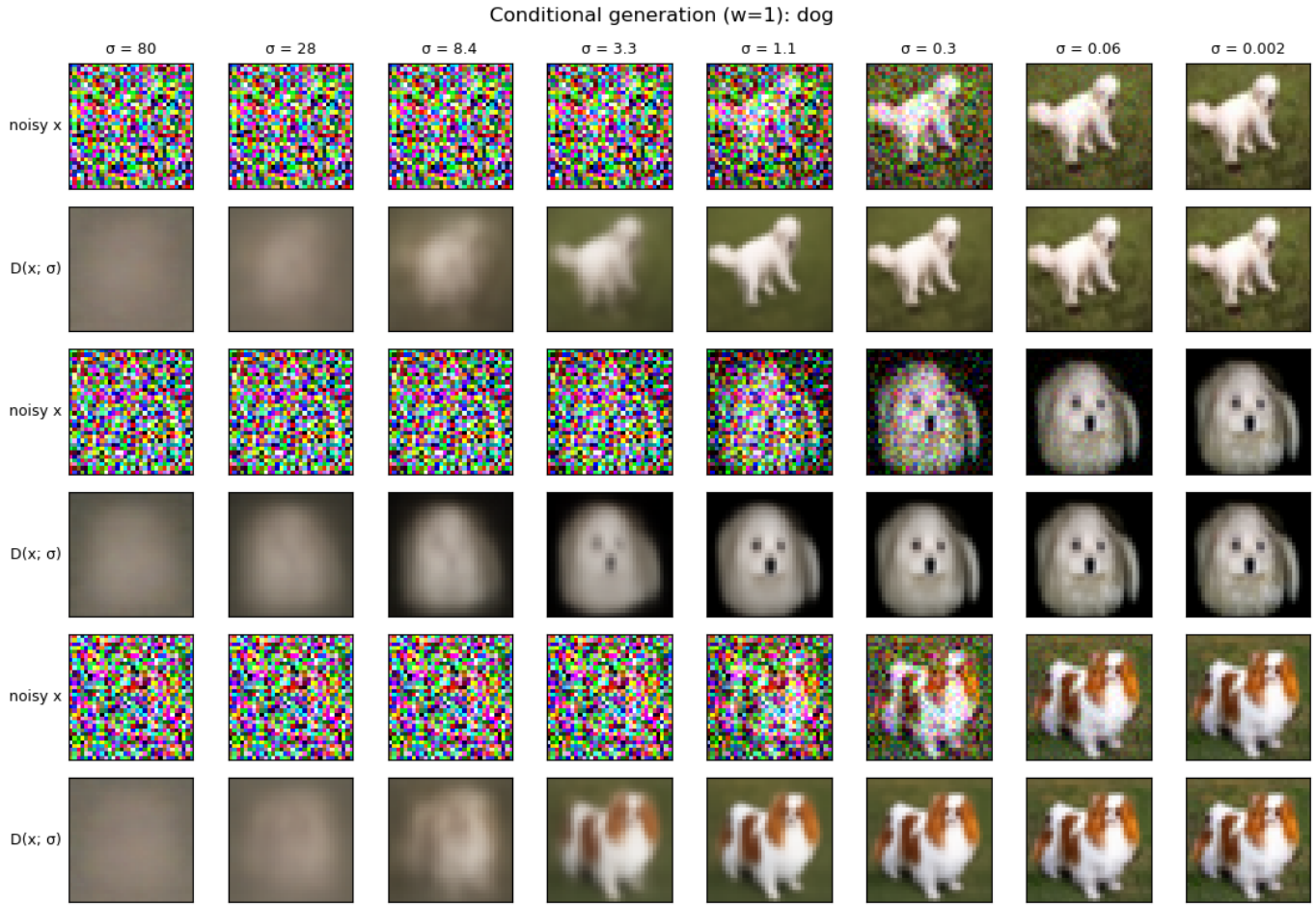


Figure 4: Sampling trajectory of the pretrained conditional model ($w = 1$) for three images requesting the class *dog* (seed 0, the same starting noise as Figure 2). Upper rows show the noisy image x and lower rows show the estimate $D(x;\sigma,c)$, at the same noise levels as before.

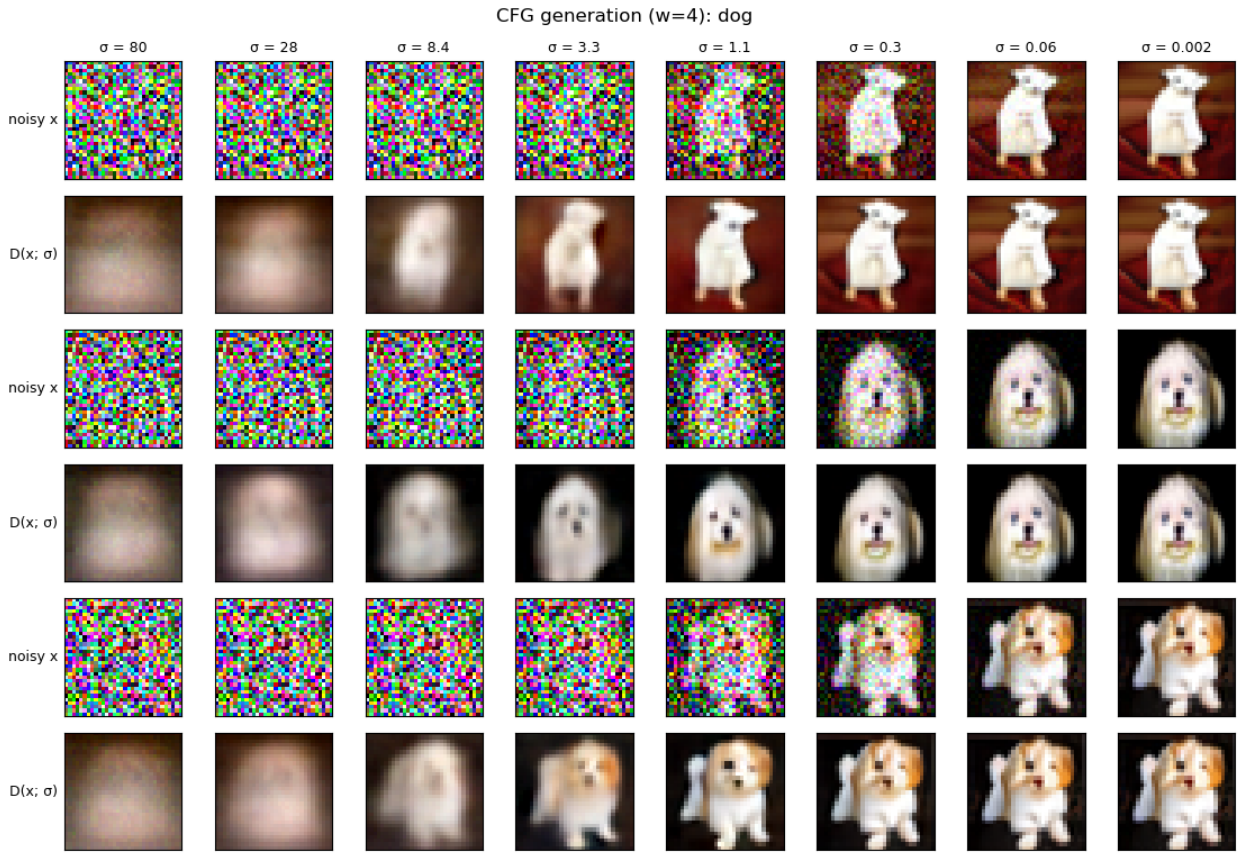


Figure 5: Sampling trajectory with classifier-free guidance at $w = 4$ for three images requesting the class *dog* (seed 0, the same starting noise as Figure 4). Upper rows show the noisy image x and lower rows show the guided estimate $D(x;\sigma) + 4(D(x;\sigma,c) - D(x;\sigma))$, at the same noise levels as before.

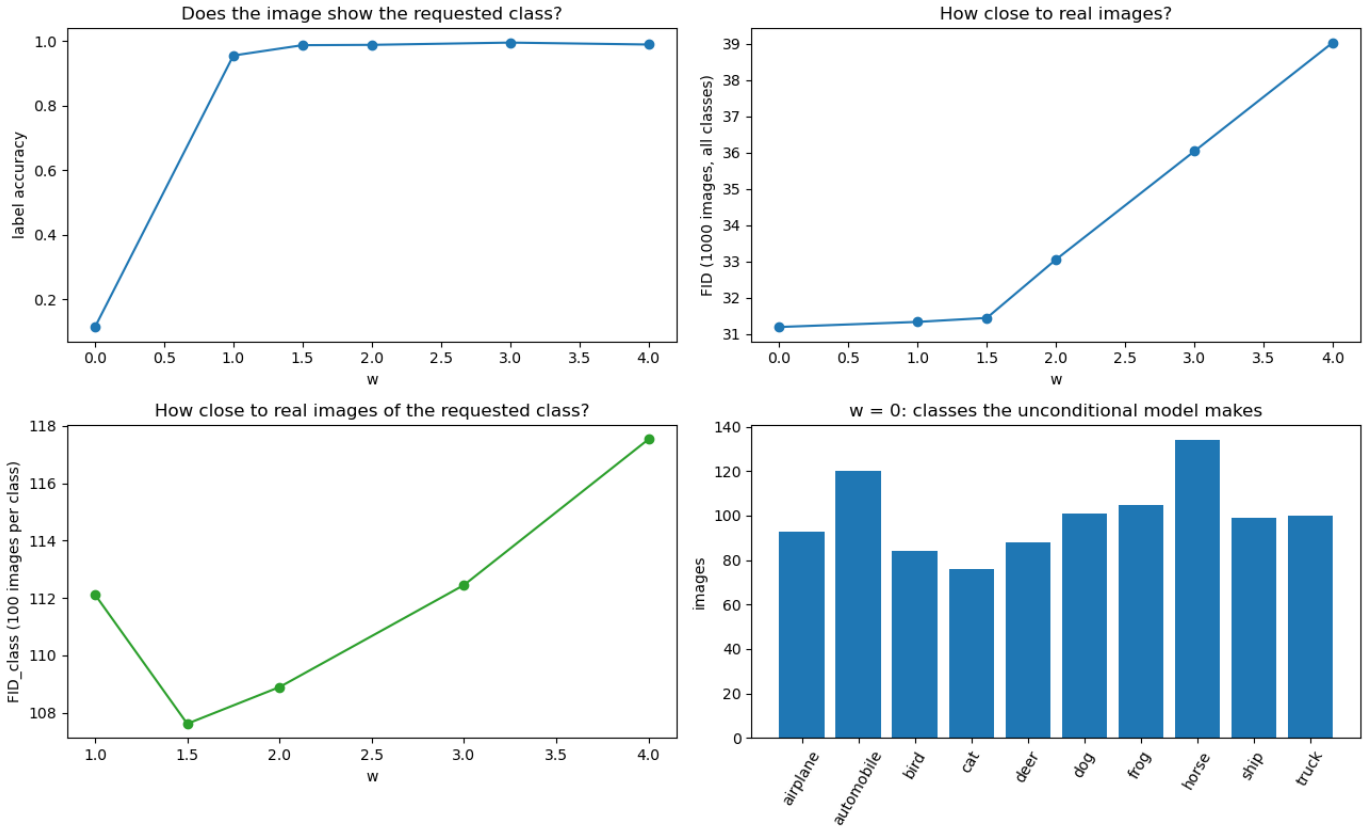


Figure 6: CFG sweep with the two pretrained networks, using 1,000 images per w with balanced requested classes. Top left: label accuracy from a ResNet-56 classifier, which rises from about 11% at $w = 0$ to about 95% at $w = 1$ and about 99% for $w \geq 1.5$. Top right: FID of all 1,000 images against the 50,000 CIFAR-10 training images, which rises from about 31 to 39. Bottom left: $\text{FID}_{\text{class}}$ (100 images per class against that class's real images, averaged over classes), lowest at $w = 1.5$; there is no $w = 0$ point because that setting ignores the requested class. Bottom right: classifier-predicted classes of the 1,000 unconditional ($w = 0$) images. FIDs from 1,000 samples are much higher than published values and are comparable only with each other.

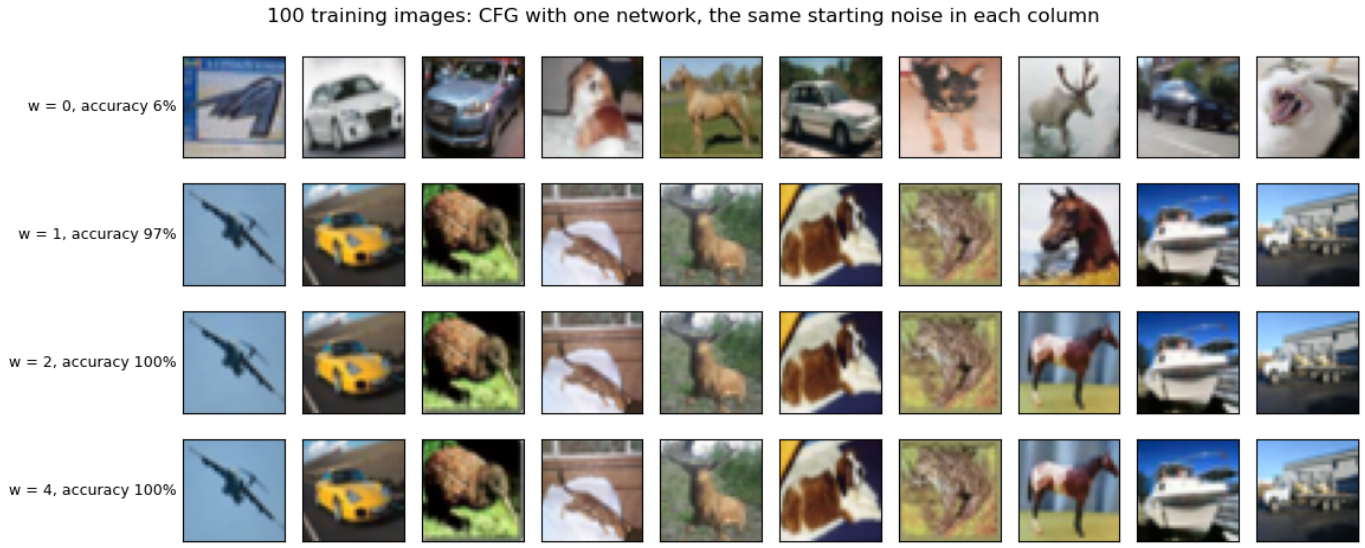


Figure 7: CFG with a single network trained from scratch on 100 CIFAR-10 images (label dropout 0.1, 7.0M parameters, 20,000 updates). Columns request classes 0–9 in order and share the same starting noise; rows show $w \in \{0, 1, 2, 4\}$, with label accuracy on 100 samples in each row label.

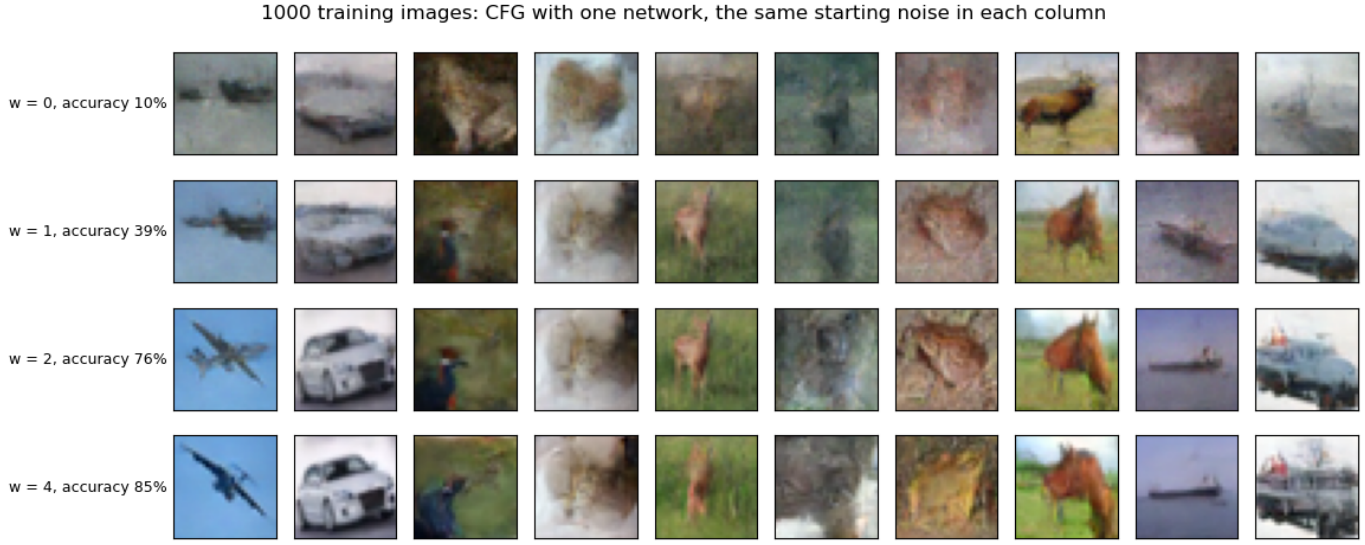


Figure 8: CFG with a single network trained from scratch on 1,000 CIFAR-10 images (label dropout 0.1, 7.0M parameters, 20,000 updates). Columns request classes 0–9 in order and share the same starting noise; rows show $w \in \{0, 1, 2, 4\}$. Label accuracy on 100 samples rises from 10% at $w = 0$ to 39%, 76% and 85% at $w = 1, 2, 4$.

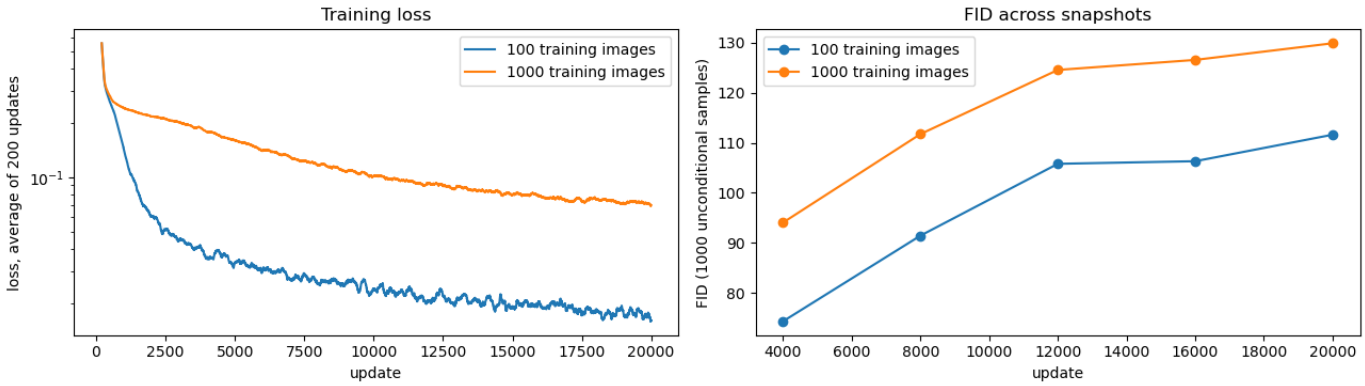


Figure 9: Training the small EDM network on 100 and 1,000 CIFAR-10 images. Left: training loss (average over 200 updates, log scale). Right: FID of 1,000 unconditional samples from the EMA weights at each snapshot, against the 50,000-image CIFAR-10 reference. The loss decreases throughout, while FID rises with training and is higher for the 1,000-image run than for the 100-image run.

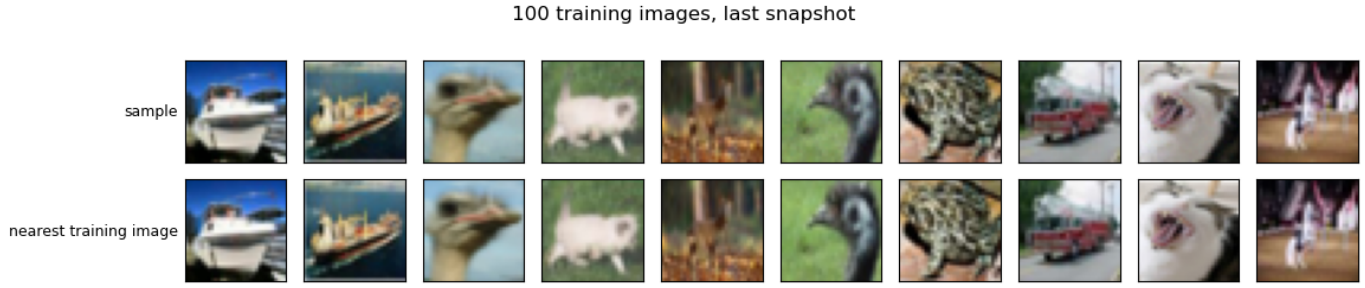


Figure 10: Unconditional samples from the final snapshot (update 20,000) of the network trained on 100 images (top row), each above its nearest training image in pixel distance (bottom row).

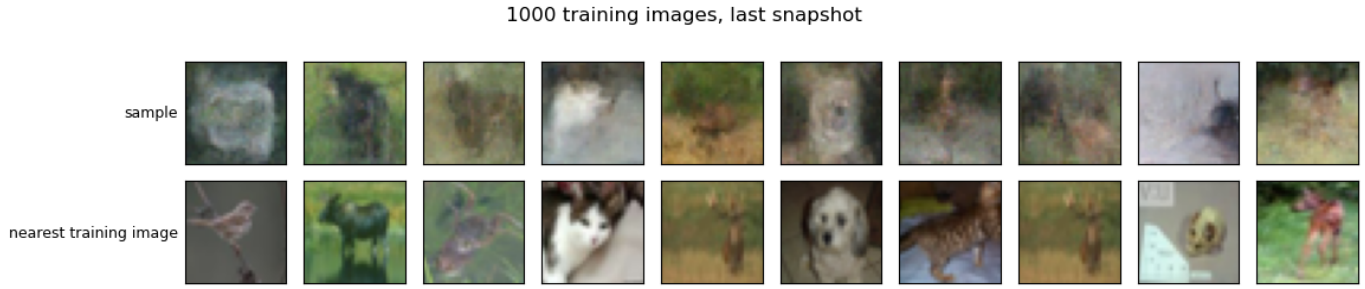


Figure 11: Unconditional samples from the final snapshot (update 20,000) of the network trained on 1,000 images (top row), each above its nearest training image in pixel distance (bottom row). The samples resemble their neighbors in layout and color but are not close copies.

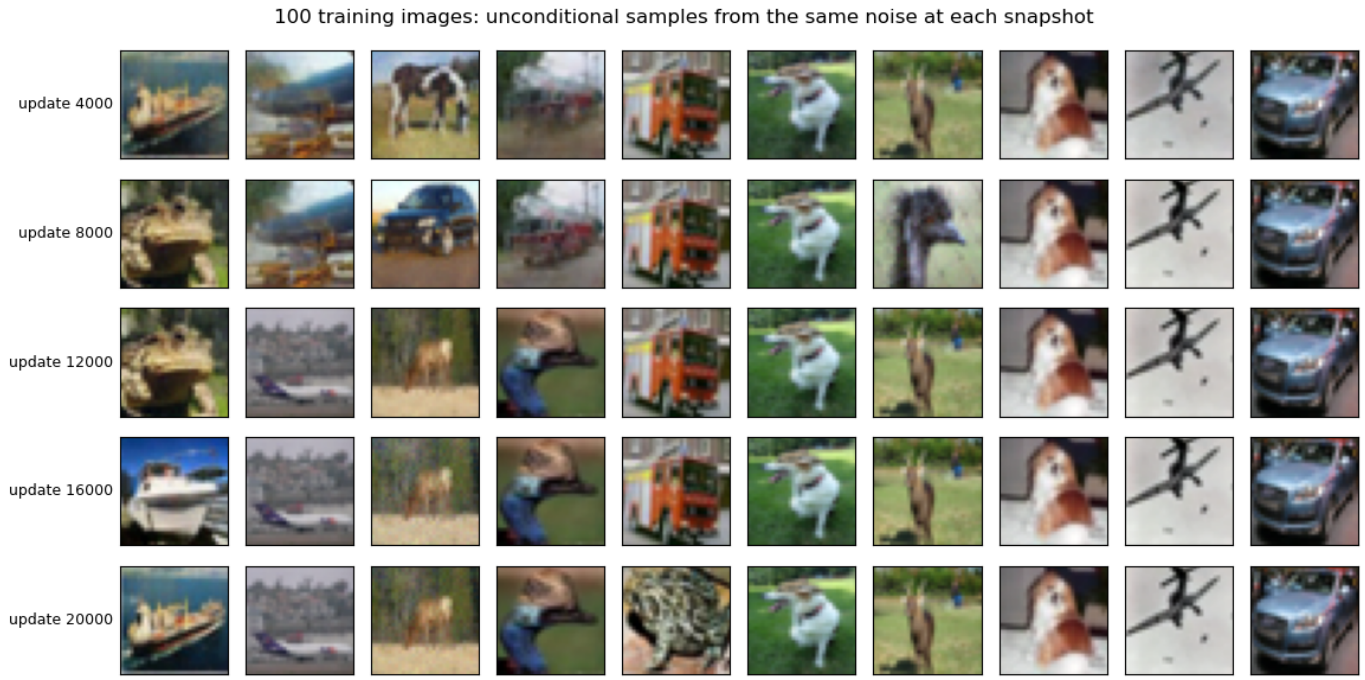


Figure 12: Unconditional samples from the network trained on 100 images at updates 4,000 to 20,000 (one row per snapshot). Each column starts from the same noise.

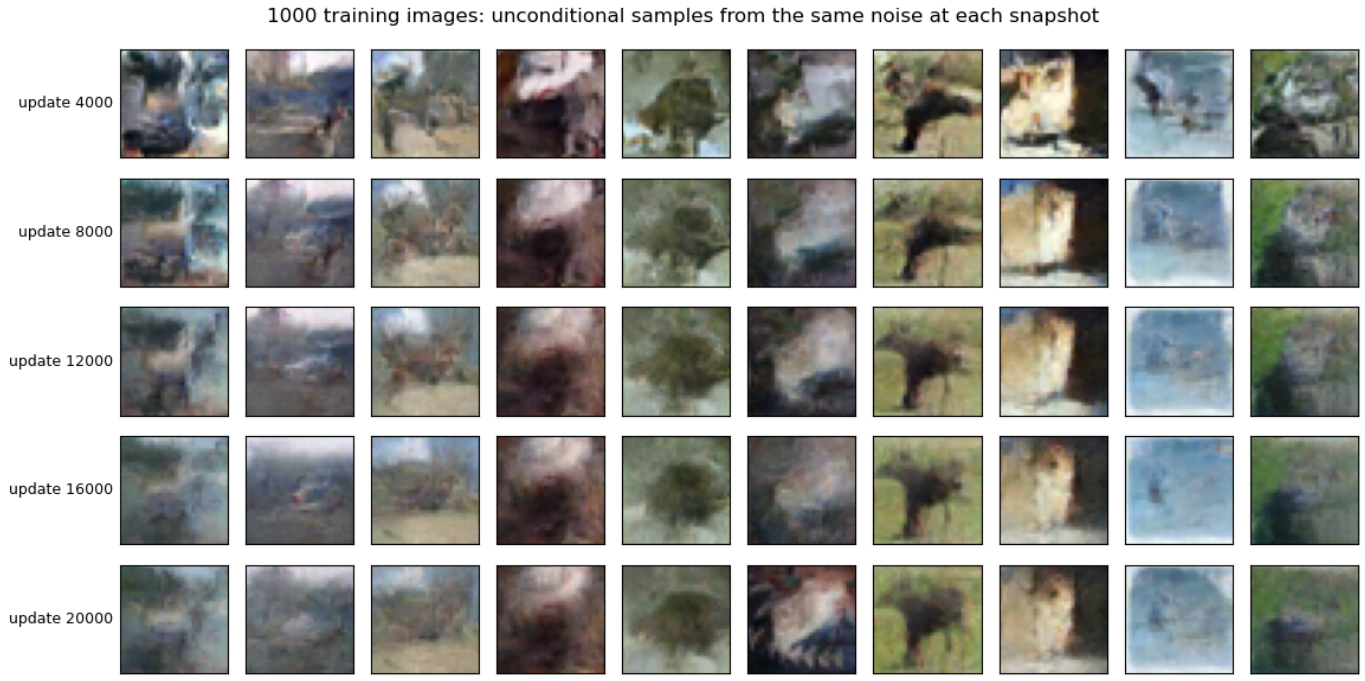


Figure 13: Unconditional samples from the network trained on 1,000 images at updates 4,000 to 20,000 (one row per snapshot). Each column starts from the same noise.

3 Results

4 Discussion

5 Conclusion

References

[1] **Carlini, Nicholas, Jamie Hayes, Milad Nasr, Matthew Jagielski, Vikash Sehwag, Florian Tramèr, Borja Balle, Daphne Ippolito, and Eric Wallace.** 2023. "Extracting Training Data from Diffusion Models." In *32nd USENIX Security Symposium (USENIX Security 23)*. USENIX Association

[2] **Dhariwal, Prafulla, and Alex Nichol.** 2021. "Diffusion Models Beat GANs on Image Synthesis." In *Advances in Neural Information Processing Systems*.

[3] **Heusel, Martin, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter.** 2017. "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium." In *Advances in Neural Information Processing Systems*.

[4] **Ho, Jonathan, Ajay Jain, and Pieter Abbeel.** 2020. "Denoising Diffusion Probabilistic Models." In *Advances in Neural Information Processing Systems*.

[5] **Ho, Jonathan, and Tim Salimans.** 2022. "Classifier-Free Diffusion Guidance." [Link]

[6] **Karras, Tero, Miika Aittala, Timo Aila, and Samuli Laine.** 2022. "Elucidating the Design Space of Diffusion-Based Generative Models." In *Advances in Neural Information Processing Systems*.

[7] **Sohl-Dickstein, Jascha, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli.** 2015. "Deep Unsupervised Learning using Nonequilibrium Thermodynamics." In *Proceedings of the 32nd International Conference on Machine Learning*.

[8] **Song, Yang, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole.** 2021. "Score-Based Generative Modeling through Stochastic Differential Equations." In *International Conference on Learning Representations*.

[9] **Wang, Qingsong, Mikhail Belkin, and Yusu Wang.** 2026. "General and Efficient Steering of Unconditional Diffusion Models." In *Proceedings of the 43rd International Conference on Machine Learning*. PMLR. [Link]

Appendices

A.1 Training Details A1
A.2 Additional Figures A1
A.3 Additional Tables A1

A.1 Training Details

A.2 Additional Figures

A.3 Additional Tables